

Internship Network in the Mathematical Sciences: Year 1 (2020-2021) Evaluation Report

Introduction

The Internship Network in the Mathematical Sciences (Inmas), a three-year project funded by the National Science Foundation (NSF), is co-directed by Fadil Santosa (Johns Hopkins University) and Richard Laugesen (University of Illinois, Urbana-Champaign). The goals of this project are to:

- Provide high quality training and internship experiences that broaden career opportunities for students in the Mathematical Sciences.
- Bring powerful computational and modeling tools to bear on industry's most pressing problems.
- Build mutually beneficial connections between industries and mathematics departments.
- Support systems change within mathematics departments in support of expanded opportunities for students and industry partners.

The major activities planned to accomplish these goals are the following.

- Launch regional networks, with hubs at Johns Hopkins University and the University of Illinois and nodes at graduate level institutions in neighboring states.
- Prepare students for the internships through technical and professional skills training.
- Seek out challenges in industry and government organizations that can be tackled by Mathematical Sciences techniques.
- Create internship projects at those host organizations, and match with students in the network.

This project began during the unprecedented Covid-19 pandemic, which required activities to shift to virtual platforms. Students selected to participate in Inmas attended four technical skills weekend workshops in Python Programming, Statistical Learning using R, Data Science and Machine Learning, Optimization and Modeling. Due to the pandemic, those were all offered via Zoom for one weekend a month, January-April 2021. The professional development skills training included workshops on resume development and behavioral interviewing via Zoom and online training modules. In addition, most of the internships were also conducted remotely.

In the first year of the program, 41 of the 101 students who applied to Inmas were accepted. Twenty-nine students completed the technical and professional development training workshops and 23 of those students were placed in internships. Of the 12 students who did not complete technical training, six completed some of the training but withdrew early after they had arranged their own internship independently; four withdrew because of time constraints; and two withdrew due to lack of sufficient technical preparation. Two additional students who did not attend the technical workshops were also placed in Inmas internships.

In selecting companies to participate and host a graduate mathematics intern, Inmas sought companies that had projects involving a high level of intellectual sophistication including some amount of mathematical or statistical reasoning. In particular, Inmas looked for places where mathematically-trained talent could make a difference and where the company did not have a history of engaging mathematical scientists. In the Mid-Atlantic, small-to-medium sized companies were targeted because we wanted to maximize potential impact to the company. The methods by which candidate host companies were identified included:

- recommendations from node university personnel in academic units and corporate relations offices,
- companies with whom the PIs have had contact in previous years,
- companies identified as holding federal Small Business Innovation Research (SBIR) grants, and
- companies located in university-affiliated research parks/business incubators.

Companies that seemed promising were then contacted by email. If they responded positively, a Zoom call was arranged with their staff to understand better their business challenges, so that we could help identify a suitable project. (Note: since the Midwest branch of Inmas was operating without a Project Developer during this stage, the companies needed to be able to identify a suitable project themselves. In the Mid-Atlantic, where a Project Developer had been hired, there was more input from Inmas in finding suitable projects.)

After learning about the company's priorities and the type of skills required for a project, Inmas identified students with relevant interests and skills for the position. Those students' resumes were forwarded to the company for consideration. We would start with students from the nearest node university. If no students from that node or from the Math Alliance were suitable for the particular project, we would draw on students from further afield in the network. This was feasible since most internships were remote, in 2021.

Evaluation Design

The adaptations that were necessary due to Covid affected the programing, instruction, student experience, and internship opportunities. The goal of the evaluation was to track successes and challenges along with those adaptations, and provide the program directors with data, observations and recommendations for continued improvements. Therefore, the first-year evaluation activities were guided by a formative approach in support of implementation and program improvement, with a primary focus on student and industry partner experiences. The second year of the evaluation will include an expanded examination of mathematics faculty and department experience. The following three overarching evaluation questions guided the first-year inquiry.

1. How did students, TAs, and instructors experience the Inmas program activities?
2. To what extent did the technical and professional development trainings affect students' awareness, skills, and confidence?
3. How did students and industry partners experience the internship matching process including preparation, opportunities, and selection?
4. What was the overall experience of the internship including fit of student skill/interest/knowledge with industry and benefits to students and industry partners?

Evaluation Activities

The evaluation activities for the first year included the following:

- An online pre-training survey link was emailed to students in February 2021¹ to learn about their technical and professional skill levels as well as perceptions about industry opportunities. All 39 participating students completed the pre-training survey.
- In April 2021, a post-training survey link was emailed to students; 26 students completed that survey.
- In May and June 2021—following the workshops and just prior to the internships—interviews were conducted with 11 students, 9 workshop teaching assistants (TAs), and 4 workshop instructors. The interviews with students included some students who withdrew early, some who obtained their own internship, and some who were placed in Inmas-sponsored internships.
- In August 2021, 4 students and 5 industry partners were interviewed to learn about the internship experience.

Survey data included descriptive statistics as well as qualitative responses to the open-ended questions. All interviews were conducted via Zoom and were recorded with participants' permission. Interview recordings were transcribed and uploaded into NVivo, a program that supports qualitative analysis. Direct quotations from the interviews are used to illustrate thematic points throughout this report. At each round of data collection, findings were shared with Inmas staff to support ongoing program improvements.

Findings

The evaluation findings followed by supporting evidence from the data are presented in four topic areas: technical workshops, professional development, internship experiences, and initial overall program outcomes.

Technical Workshops

Four technical training workshops were conducted via Zoom one weekend a month during January-April 2021. The topics covered were: Python Programming, Statistical Learning using R, Data Science and Machine Learning, Optimization and Modeling. Individual instructors were contracted by Inmas to conduct each workshop with two or three TAs assigned to each.

Key Findings Based on Survey and Interview Data

Implementation

- Most everyone (students, TAs, and instructors) commented that an online experience for learning was not ideal; however, they thought it went as well as it could have given the pandemic circumstances.
- Students, TAs, and instructors all noted that sharing clear expectations about zoom etiquette and participation in breakout rooms at the beginning of the program—along with timely reminders—would be beneficial.
- TAs and instructors shared that more communication from Inmas staff about the overall structure of the program would be helpful.

¹ The survey was administered after the first workshop due to contractual constraints.

- TAs observed that it might be more advantageous to stay with one language for all four workshops, and given that Python is so widely used in industry, that might be the best choice.
- Instructors and TAs reported that they thought communication and interactions among TAs and instructors went very well.
- Students reported little or no interactions with other students outside of the workshop; a sense of community was challenging to support or sustain in an online environment.
- Students, TAs, and instructors all noted the positive impact the participation of the Inmas co-directors had on the workshop experience.

Student Gains in Technical Skills and Confidence

- Students had varied levels of experience prior to the workshops ranging from no experience at all to proficiency. Post-training survey responses show a growth in skills in all four areas.
- Students who had no prior experience in either programming language (Python or R) reported struggling to absorb what was being taught during the weekends; whereas, students who had some experience in one language found it easier to learn the other language.
- At the end of the workshops, students reported confidence in their skills in the four areas ranging from not at all confident to proficient. This range was affected by students' prior skill level and their comfort with proceeding with a few tools to approach a new skill.

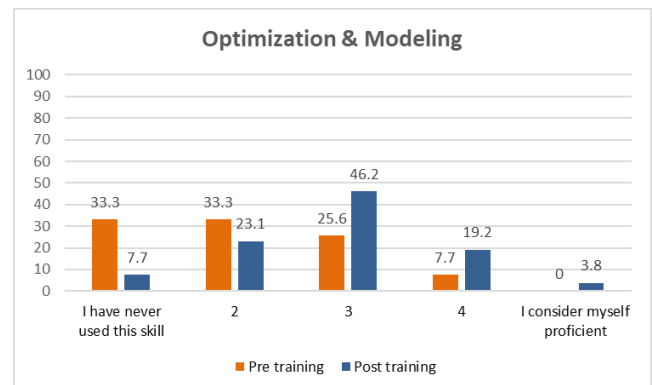
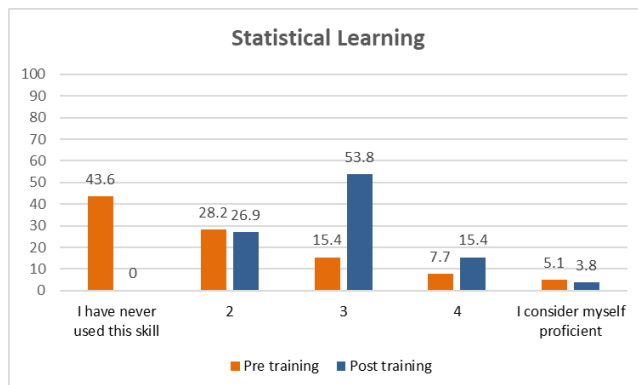
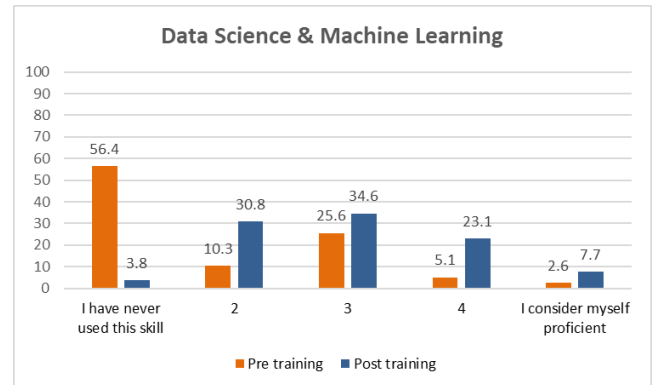
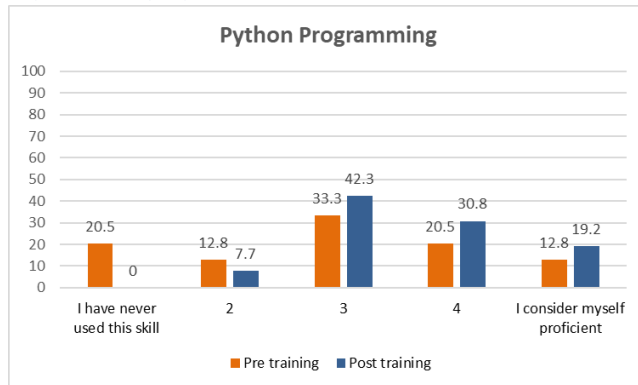
Industry Partner Feedback on Student Technical Skills

- Industry partners who hired interns were pleased to learn about the workshops and reported that the interns learned quickly and were competent and skilled in their contributions during the internships.

Supporting Evidence for Technical Workshop Findings

The following sections present survey results and selected quotations from interviews in support of the finding on the technical training workshops.

Pre- and Post-Training Survey Results: How would you rate your proficiency in the following skills? (Percentages)



Student Interview Quotations

- "I think a lot of good information was given to me in a short amount of time, and I think I did what I could in the group settings but I did find myself throughout the course of the weekend just a little overwhelmed, and then frustrated not being able to pick up as much as they're throwing at me."
- "Before the training, I had no idea about these programs. But at least now I know how to install the program and I know where to ask the question on the internet. If I need to use them, I'm sure I can use both of them [Python and R]. Of course, I need to spend some time Googling something but I am sure I can do that."
- "Actually, I am quite proficient with R because that is my main programming language for my daily research because I am a statistics major in my PhD program. But the first day that workshop taught me some new skills to plot graphs. And that is a very advanced skills that I have never used before."
- "I had never used R before. So it was a little rough for me because I had never used that language. I would not feel comfortable saying to someone, 'Oh yeah, I know how to use R.' But if, for some reason, something came up and I had to learn R, I'm probably in a far better position

than I would have been before. But it's not to the place where I would put it on my resume or anything like that. I still don't feel confident at all, but I do feel like it helped if it comes up in the future."

- "I interacted with others during some of the workshops and some of the career sessions when we had small breakout rooms. That was actually kind of nice, because, with the pandemic, you don't get a lot of personal interaction. Sometimes when we were in the breakout rooms, we'd work on a programming problem for a while and there would just be a natural pause, and it would turn into, 'So how's everyone doing?' And we would talk to each other about how the intern search was coming along and things like that."
- "I was assigned to the same breakout room with [one of the co-directors] in the last workshop. We actually worked on the same Optimization problem together using Python. And, oh my God, I was so motivated by this kind of spirit because they're professors and still feel so excited about learning new things. So actually, I feel great about that, too."

TA and Instructor Quotations

- "For such a short collection of workshops, Inmas really needs to stick to one language, which is Python because that's what everyone uses in industry nowadays. And that's the language you can learn in three workshops... I mean, R is like, was historically the industry language but really in like the last 10 years, it's heavily shifted to everyone does Python." – TA
- "I thought the TAs were great. They were really active in the working times, and they helped out a lot there. And they were also really active in the lectures, and they were asking a lot of questions in the lectures, which I thought contributed to those." --Instructor
- "I find it really kind of cool that the co-directors were both present throughout the weekend. I mean, I don't think that's anything that can be taken for granted. I found this very pleasant that I was able to interact with them in this way during the weekend." --Instructor

Industry Partner Quotations

- "This was our second PhD level data science intern and he probably did as well as that PhD. He picked up everything he did really, really quickly... So that definitely makes sense that he had some workshops in it. Because he came in like somebody who knew more than we thought."
- "She also jumped into some things we didn't expect her to be quite as technical on things like notebooks and data sets and machine learning. She actually... I don't know if she knew that stuff or just learned it, but she was able to get in and, and do some really nice work for us."

Professional Development

The coaching and professional development components of Inmas, conducted by a career coaching professional, included workshops on resume development and behavioral interviewing plus self-directed online modules and an online Discord discussion group.

Key Findings Based on Survey and Interview Data

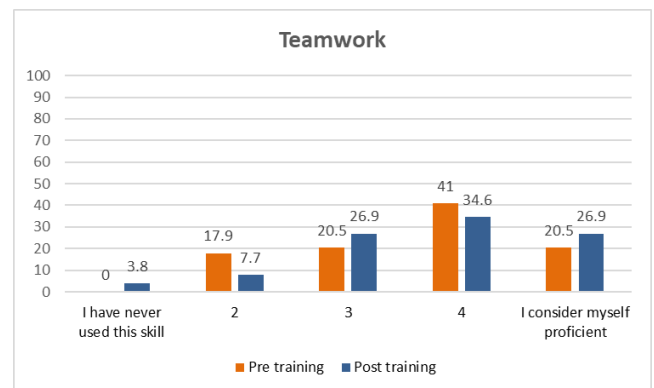
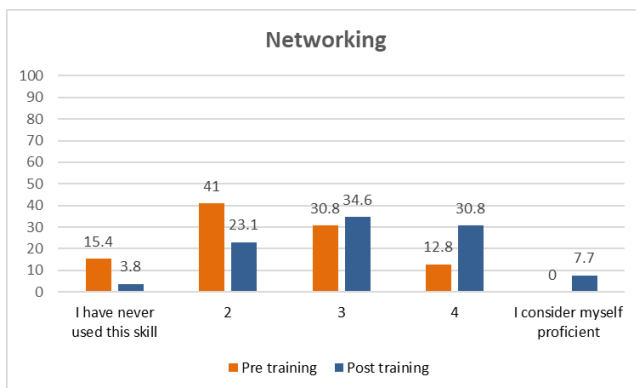
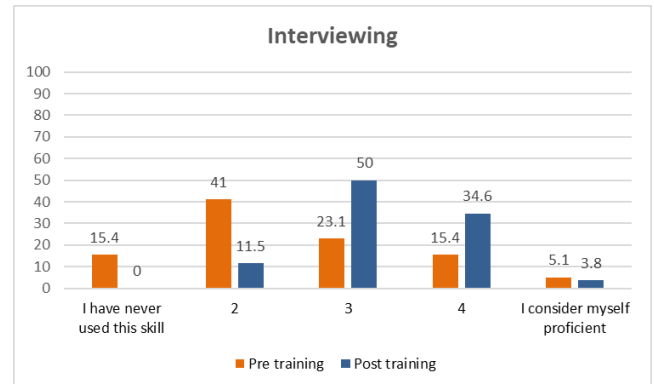
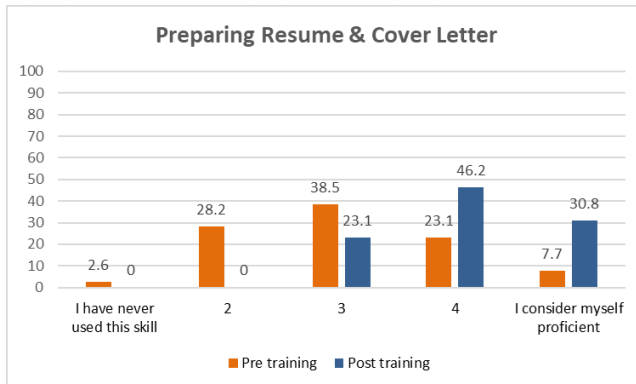
- Overall, students found the introduction and approaches to applying to industry jobs helpful. Students indicated that the resume workshop was the most useful component followed by the interviewing tips and practice. These two areas also showed the great gain in students' confidence.
- The self-directed online modules were less effective. Students reported that the number of modules felt overwhelming with the content being only modestly helpful.

- Most students indicated that they did not participate in any discussions on Discord.

Supporting Evidence for Professional Development Findings

The following sections present survey results and selected quotations from interviews in support of these finding on the professional development activities.

Pre- and Post-Training Survey Results: How would you rate your proficiency in the following skills, in a professional setting? (Percentages)



Student Quotations

- “I was unaware about so many things, like what happens in the industry sector, how people look for jobs, and what is the first part. So all these things I learned from the instructor of this professional activity.”
- “I feel like most people don't really like talking about themselves, don't really like working on their resumes, and it kind of forced you to do all of those things. It's nice that now I have a CV and a resume that are both up to date and in good shape. Now if I do something new and exciting, I just have to add it on.”
- “They had you in little breakout rooms where you would share your resume with someone, just another one of the students, and ask clarifying questions and comments and things like that. In some of those small meetings, I had people saying, ‘Oh, well that sounds cool. Why isn't that on your resume?’ I think it encouraged me to put a couple more lines on there that maybe that I didn't really feel were impressive before.”
- “The online modules felt more like, ‘Oh, I need to complete this because there's a deadline.’ And then while I was doing it, I don't think that, ‘Oh wow, that's cool.’ Or, ‘Oh, I didn't realize that before.’ I didn't have any of those moments while I was completing them.”

Internship Experience

As noted earlier, Inmas staff used criteria to select companies with the greatest opportunities to utilize and benefit from interns with high-level mathematical training. In addition, Inmas staff sought to match particular student interest and skills with the problems identified by companies.

Key Findings Based on Survey and Interview Data

- Overall, students shared that they were quite excited about their internships and thought that the process for obtaining an internship was fairly straightforward.
- Students said they appreciated the support and assistance that Inmas staff provided.
- Students noted that Inmas internships provided opportunities that they may not have known about or pursued on their own.
- Overall, companies reported that they were happy with how the intern selection process went. They felt that Inmas had done some preliminary screening so that the resumes they reviewed all seemed to fit well with their company and needs.
- Industry partners were very impressed by the caliber of the interns, often noting the unique contributions made by the intern.
- A number of industry partners were interested in extending the internships into the fall semester.
- Industry partners observed that the Inmas program provided an opportunity for graduate students that otherwise would be difficult to obtain in their industry as funding and structures within their companies do not usually support internships.

Supporting Evidence for Internship Experience

Student Quotations

- “I prepped for the interview probably harder than I've ever prepped for any other interview in undergrad. Inspired by a lot of the stories that [the professional development instructor] had about presenting yourself as somebody who knows the material beforehand, I searched out the exact papers that were being written about the project or that the project was basing network on and I read those before showing up to the interview. I demonstrated myself to be a capable candidate.”
- “I got an email from Inmas staff that said, ‘Hey, I sent your resume over. Expect to hear from them soon.’ And soon after the guy from the company contacted me about an interview. So it was pretty streamlined.”
- “This [internship experience] definitely helps me to see all the different things that I could do having a math background and I was really introduced into data science. I'm really grateful for this experience. It's really helped to see sort of where I'm headed.”

Industry Partner Quotations

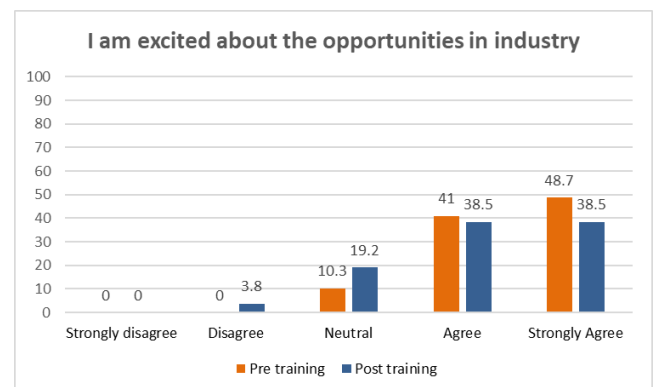
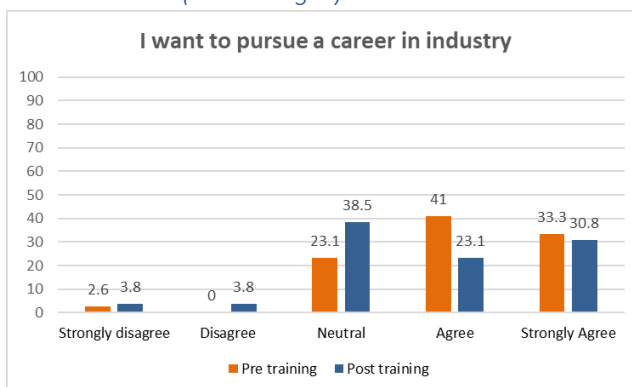
- “Everything was very smooth. The results were really incredible. I mean, we were very happy with the Inmas internship experience.”
- “I only have positive things to say and a lot of which I already expressed and maybe just to reemphasize a couple of points: One, this type of funding fills a gap that that is mutually beneficial. I'm not aware of other avenues to make this sort of thing happen. And internally, it was extremely positive that we were able to have someone focus. And then just thinking about the benefits for the intern.”

- “The Inmas intern has been developing simulation tools where you can go onto a website or go into a software and run simulations. I guess the two most important things that we saw with the project was that she took initiative to develop things that I haven't prescribed, she extended some of the ideas that I had related to the project. A couple of times she's engaged with other stakeholders that aren't directly in my department and sort of answered questions and worked with them on some problems they had.”

Overall Program Outcome

In addition to increased knowledge and confidence in the technical and professional skills discussed earlier, a key outcome the program sought to achieve was an increase in students' awareness and interest in industry career opportunities. The data obtained via the pre- and post-training surveys indicate that the weekend workshops did not significantly change students' interest or awareness; however, the interviews that were conducted following the internship experience revealed a much greater shift in perception and excitement in industry positions. While it is understandable that an on-the-ground experience would have a greater impact, the co-directors also plan to include more exposure to professionals working in industry during the technical workshops phase of the program.

Pre- and Post-Training Survey Results: How would you rate your level of agreement to the following statements? (Percentages)



Supporting Evidence for Expanded Awareness of Industry Opportunities

Student Quotations

- “Before joining this program, I was 90% towards academia and 10% towards industry. But after joining this program, I got to know several aspects of industry, like what kind of problems are there in different industries. I was surprised during one discussion to learn that there are some positions in industry where you can do research.”
- “A month ago, in our department there was a post doc who just finished his three years postdoc position in our university and moved to industry. So maybe right after the PhD, I may go to some academic positions but it does not mean that I will be in academia forever.”
- “I didn't know what working in industry was like at all. I was just making assumptions. So this program really gave me a sense of what it would actually be like so I could make an informed decision when it came time to graduating and deciding what's next. So yeah, I would say it made a big impact.”

Recommendations

The preliminary findings from the surveys and the two rounds of interviews were shared with Inmas staff as each data collection cycle was completed. As a result, program staff began to make changes well before this final report for Year 1 was submitted. Below is a brief summary of those recommended changes.

1. Inmas will focus on Python as the programming language for all technical workshops, no longer conducting the statistical learning workshop in R.
2. A different approach and vendor will be used in Year 2 for the professional development components of the program (i.e., resumes and interviewing).
3. In addition to the current professional development opportunities, Inmas will provide an introduction to corporate teamwork approaches such as Agile.
4. The program will provide more opportunities to hear from mathematicians who are in industry, especially during the technical workshop phase of the program.
5. Instructors, TAs, industry partners, and students shared that they would have benefited from more information about the overall Inmas vision and background. Inmas staff will create communication materials to help fill those gaps.